

Machine Learning

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Team Project Assignment

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Team Project Assignment

All teams implement the same machine learning specializations on their own assigned dataset/domain. Each team has been assigned a unique **Problem Domain** (housing prediction, churn prediction, image classification, NLP sentiment, game environments, anomaly detection, etc.) and will progressively implement all six learning approaches within that domain across different workshops. This ensures comprehensive coverage of modern ML techniques while maintaining domain-specific expertise.

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Detailed Topic Angle Descriptions

Each team is assigned a unique problem domain and will implement all six ML specializations within that domain across successive workshops:

1. **Team 1 — Housing Price Prediction:** Domain: Real estate valuation. Dataset: Ames Housing or Boston Housing. Implementations across workshops: (W1) Supervised regression models (linear, tree-based, ensemble); (W2) Deep learning with neural networks; (W3) Reinforcement learning for investment decisions; (W4) Autoencoders for feature learning; (W5) GANs for synthetic housing data; (W6) Medical image processing adaptation for architectural analysis.
2. **Team 2 — Customer Churn Prediction:** Domain: Telecommunications/banking customer retention. Dataset: Telco Customer Churn or similar. Implementations: (W1) Supervised classification; (W2) Deep neural networks for sequential patterns; (W3) RL for retention strategy optimization; (W4) Autoencoders for customer profile learning; (W5) GANs for synthetic customer generation; (W6) Medical domain transfer learning.
3. **Team 3 — Image Classification (CIFAR-10/MNIST):** Domain: Computer vision and object recognition. Dataset: CIFAR-10, MNIST, or Fashion-MNIST. Implementations: (W1) Supervised classification (traditional ML); (W2) CNNs and deep convolutional architectures; (W3) RL for image navigation or active learning; (W4) Autoencoders for dimensionality reduction; (W5) GANs for image generation; (W6) Medical image classification adaptation.

4. **Team 4 — Sentiment Analysis (NLP):** Domain: Natural language processing and text classification. Dataset: IMDb reviews, movie ratings, or product reviews. Implementations: (W1) Supervised text classification; (W2) RNNs/LSTMs and transformer models; (W3) RL for dialogue systems; (W4) Autoencoders for text representation; (W5) GANs for text generation; (W6) Medical text mining and diagnosis support.
5. **Team 5 — Game Playing (RL Environments):** Domain: Reinforcement learning and game AI. Environments: CartPole, LunarLander, MountainCar, or Atari. Implementations: (W1) Supervised learning baselines; (W2) Deep Q-Networks and value-based methods; (W3) Policy gradient and actor-critic algorithms; (W4) Autoencoders for state representation; (W5) GANs for adversarial agents; (W6) Medical robotics control adaptation.
6. **Team 6 — Anomaly Detection (Time Series):** Domain: Outlier detection and system monitoring. Dataset: Credit Card Fraud, ECG heartbeat, network traffic, or sensor data. Implementations: (W1) Supervised classification of anomalies; (W2) Deep neural networks for sequential anomaly detection; (W3) RL for adaptive thresholds; (W4) Autoencoders and VAEs for unsupervised anomaly detection; (W5) GANs for synthetic normal/anomalous data; (W6) Medical ECG and vital signs anomaly detection.
7. **Team 7 — Recommendation Systems:** Domain: Collaborative filtering and personalization. Dataset: MovieLens, user-item ratings, or e-commerce. Implementations: (W1) Supervised matrix factorization; (W2) Deep collaborative filtering with neural networks; (W3) RL for sequential recommendation; (W4) Autoencoders for latent factor learning; (W5) GANs for synthetic user-item interactions; (W6) Medical treatment recommendation systems.
8. **Team 8 — Text Generation and Language Models:** Domain: Generative NLP and sequence generation. Dataset: Shakespeare, news articles, or open-source corpora. Implementations: (W1) Supervised sequence-to-sequence models; (W2) Transformer-based language models; (W3) RL for controllable text generation; (W4) Autoencoders for semantic compression; (W5) GANs for adversarial text generation; (W6) Medical report and diagnosis generation.

Workshop-Based Implementation Progression

All teams progress through the same specializations in each workshop, but applied to their unique domain:

- **Workshop #1 — Supervised Learning:** Data exploration and EDA, dataset preparation, feature engineering, implement regression/classification algorithms (linear models, decision trees, ensemble methods), hyperparameter tuning, evaluation with domain-specific metrics
- **Workshop #2 — Deep Learning Basics:** Build neural network architectures (MLPs, CNNs for images, RNNs/LSTMs for sequences), implement training pipelines with TensorFlow/PyTorch, apply regularization and optimization techniques, generate learning curves and performance analysis

- **Workshop #3 — Reinforcement Learning:** Design reward functions and environment interactions relevant to domain, implement Q-learning or policy gradient methods, train agents, analyze convergence and policy behavior, visualize decision-making
- **Workshop #4 — Unsupervised Learning (Autoencoders):** Implement autoencoders and VAEs, apply for dimensionality reduction or anomaly detection, visualize latent space representations, evaluate reconstruction quality and generative properties
- **Workshop #5 — Generative Adversarial Networks (GANs):** Design generator and discriminator networks, implement training procedures with stability techniques, generate synthetic domain data, evaluate generation quality, analyze mode diversity and convergence
- **Workshop #6 — Medical Image Processing Adaptation:** Apply learned techniques to medical imaging tasks, implement domain transfer strategies, handle medical data specifics, apply interpretability methods (saliency maps, attention), evaluate clinical relevance metrics

Comprehensive Delivery Schedule

- **Initial Setup:** Domain selection, dataset acquisition and validation, exploratory data analysis report, project roadmap documentation
- **Workshop #1 (Supervised Learning):** Supervised model implementations (regression/classification), feature engineering report, baseline performance metrics, comparison of algorithms
- **Workshop #2 (Deep Learning):** Neural network implementations, architecture design documentation, training curves analysis, performance benchmarking against Workshop #1
- **Workshop #3 (Reinforcement Learning):** RL environment setup, policy implementation, training dynamics analysis, reward function justification, convergence visualization
- **Workshop #4 (Autoencoders):** Autoencoder/VAE implementations, latent space analysis, dimensionality reduction results, anomaly detection or generative quality assessment
- **Workshop #5 (GANs):** GAN architectures (basic GAN, DCGAN, WGAN), synthetic data generation, quality metrics (FID, IS, visual assessment), training stability analysis
- **Workshop #6 (Medical Adaptation):** Domain transfer and medical application implementation, clinical metric evaluation, interpretability analysis, comparative results across all specializations